

Data Cleaning & Reporting Automation Using Python

Aim:

To automate data cleaning and reporting workflows using Python by handling missing values, duplicates, outliers, and generating summary reports.

Objectives:

- To identify and handle missing values in the dataset
- To detect and remove duplicate records.
- To identify outliers using statistical techniques.
- To automate data cleaning processes using Python.
- To generate automated summary reports and visualizations.
- To improve data quality for effective analysis and decision-making.

Tools used:

- Python
- Pandas
- Matplotlib
- Jupyter Notebook

Dataset Used:

Mall CustomersDataset

Visualizations:

1) Dataset preview

[1]:	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

2) Dataset Information

```

Shape: (200, 5)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   CustomerID                           200 non-null    int64
1   Gender                               200 non-null    object
2   Age                                   200 non-null    int64
3   Annual Income (k$)                   200 non-null    int64
4   Spending Score (1-100)                200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB

```

3) Missing Value Analysis

```

CustomerID      0
Gender           0
Age             0
Annual Income (k$)  0
Spending Score (1-100)  0
dtype: int64

```

4) Duplicate record analysis

```
Duplicate Records: 0
```

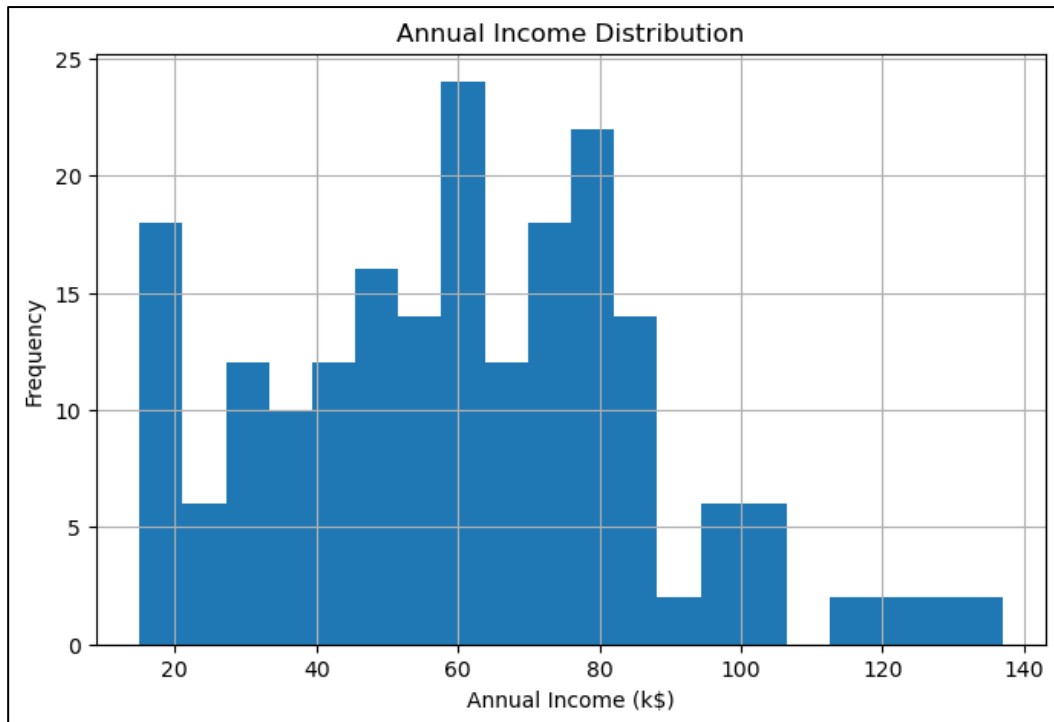
5) Outlier detection

```
Number of Outliers: 2
```

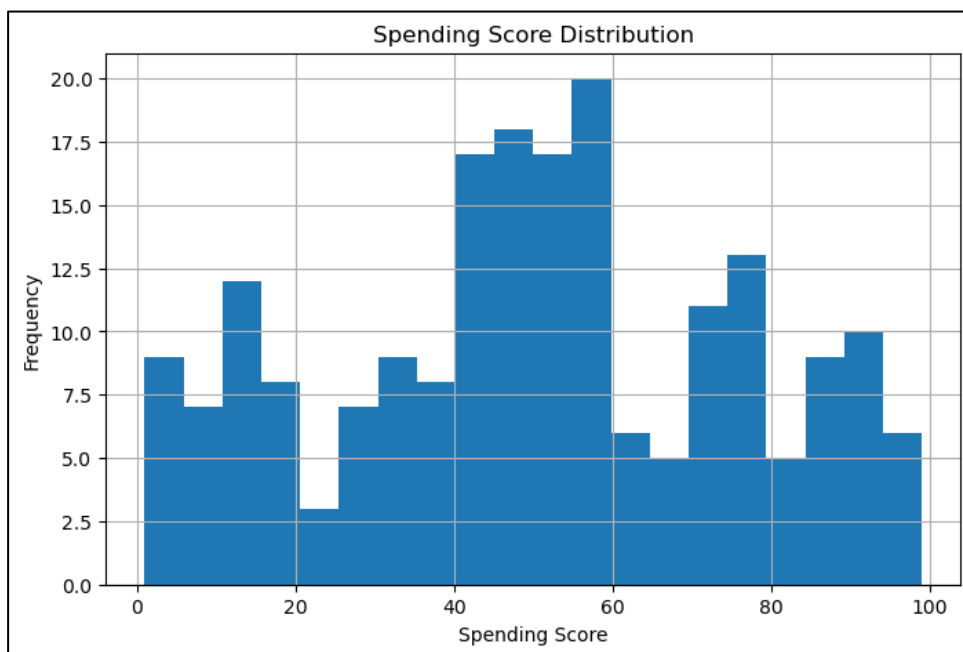
6) Summary statistics report

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

7) Annual income distribution



8) Spending score distribution



9) Automated report generation

Automated Report Generated Successfully

	A	B	C	D	E	F	G	H
1		CustomerID	Age	Annual Inco	Spending Score (1-100)			
2	count	200	200	200	200			
3	mean	100.5	38.85	60.56	50.2			
4	std	57.8791845	13.9690073	26.2647211	25.823521668370162			
5	min	1	18	15	1			
6	25%	50.75	28.75	41.5	34.75			
7	50%	100.5	36	61.5	50			
8	75%	150.25	49	78	73			
9	max	200	70	137	99			
10								
11								
12								

Key findings:

1. The dataset was successfully inspected for missing values and duplicates
2. Outliers were identified using the IQR method.
3. Annual income and spending score distributions provided useful customer insights.
4. Automated report generation reduced manual reporting effort and improved efficiency.

Conclusion:

The project successfully automated data cleaning and reporting tasks using Python. Missing values, duplicate records, and outliers were handled efficiently, while automated reports and visualizations enhanced data analysis and reporting productivity.